

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10: 1.4 ENERGY, WORK, POWER AND MACHINES

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Final Explanation: How do pulleys reduce the effort needed to lift water, and how does energy transfer in the process? Imagine you are explaining to a friend in Kitui County why Mwende, who is 10 years old, can lift the bucket with the pulley system while her mother, Mama Kavata Mutua, finds the direct lift exhausting. Use what you observed and learned in Lessons 1 to 8. Use $g = 10 \text{ N/kg}$ everywhere. Show every calculation step, with units, and write full sentences for your explanations. Use the data given in each section. Write your answers in the space under each prompt.

Part 1: What the bucket gains

A 20 kg bucket of water (weight 200 N) is lifted 8 m up a well at a steady speed in 20 s. The water surface is the 0 m level and the top of the well is 8 m above it.

Bucket PE table ($m = 20 \text{ kg}$):

Height above water (m)	PE (J)
0	0
2	400
4	800
6	1200
8	1600

(a) Use $PE = mgh$ to show how

(a) $PE = mgh = 20 \text{ kg} \times 10 \text{ N/kg} \times 8 \text{ m} = 1600 \text{ J}$. (Equivalently, weight $200 \text{ N} \times 8 \text{ m} = 1600 \text{ J}$.)

(b) The bucket gains exactly 1600 J of potential energy in both cases. PE depends only on the weight of the bucket (200 N) and the height it is raised (8 m). Both ways raise the same weight through the same height, so the energy gain is the same. The pulley system does not give the bucket extra energy.

(c) The bucket moves at a steady speed, so its kinetic energy does not change. The whole energy gain of 1600 J is gravitational potential energy.

the table value for 8 m is found. (b) Mama Kavata lifts the bucket by hand. Mwende lifts it with the pulley system. Compare the energy the bucket gains in the two cases and explain why. (c) The bucket moves at a steady speed. What happens to its kinetic energy, and where does all of the energy gain go?	
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Part 2: Work, force and distance

Use $W = Fd$, with F in the direction of motion. (a) Calculate the work Mama Kavata does lifting the bucket directly by hand through 8 m, and her power if the lift takes 20 s. (b) With the pulley system, Mwende pulls the rope with an effort of 125 N and the rope is pulled 16 m while the bucket rises 8 m in 20 s. Calculate the work she does and her input power. (c) Compare the force and the distance in the hand lift and the pulley lift. What does the pulley system change, and what does it not change about the work needed?	(a) $W = Fd = 200 \text{ N} \times 8 \text{ m} = 1600 \text{ J}$. Power $P = W/t = 1600 \text{ J} / 20 \text{ s} = 80 \text{ W}$. (b) Work input = $125 \text{ N} \times 16 \text{ m} = 2000 \text{ J}$. Input power = $2000 \text{ J} / 20 \text{ s} = 100 \text{ W}$. (c) The pulley system lets Mwende pull with a smaller force (125 N instead of 200 N), but over a longer distance (16 m instead of 8 m). The pulley does not reduce the work needed: a machine trades a smaller force for a longer distance. In fact the work she does with the pulley system (2000 J) is more than the 1600 J of the direct lift, because of friction in the pulleys and rope.
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Part 3: Where the energy comes from and where it goes

Use conservation of energy. (a) Where does the energy come from to lift the bucket,	(a) The energy comes from the chemical energy in the person's body, from food such as ugali and githeri. The pulley cannot create energy, because energy is never created or destroyed, only transformed or transferred. (b) $2000 \text{ J} - 1600 \text{ J} = 400 \text{ J}$. This 400 J is transferred as heat (and a little sound) to the surroundings by friction in the pulleys and rope. It is not destroyed, so energy is conserved: $2000 \text{ J input} = 1600 \text{ J of PE} + 400 \text{ J of heat}$.
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and can the pulley create energy? Explain. (b) In the pulley lift, the bucket gains 1600 J of PE but Mwende does 2000 J of work. Calculate the difference and explain where this energy goes. Is it destroyed? (c) In Lesson 4, a real bucket fell from 8 m and hit the water at 11 m/s. Calculate its KE at the water using $KE = \frac{1}{2}mv^2$, and the energy transferred to the surroundings, given that its PE at the top was 1600 J. How does this help you explain your answer to (b)?	(c) $KE = \frac{1}{2} \times 20 \text{ kg} \times (11 \text{ m/s})^2 = \frac{1}{2} \times 20 \times 121 = 1210 \text{ J}$. Energy transferred = 1600 J - 1210 J = 390 J, which went to heat, sound and air/rope friction. This shows that in real systems 'lost' energy is transferred to the surroundings, not destroyed. In the same way, friction in the pulley system transfers 400 J to heat.
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Part 4: Machines, MA, VR and efficiency	
The pulley system has one fixed pulley and one movable pulley, with 2 rope sections supporting the bucket. The load is 200 N and the actual effort is 125 N. The rope is pulled 16 m while the bucket rises 8 m. (a) Calculate the MA, the VR and the efficiency of the pulley system. (b) Calculate the effort needed if the pulley system had no friction (an ideal pulley system), using the VR. (c) A crowbar lifts a 400 N load with an effort of 125 N. The effort arm is 1.2 m and the load arm is 0.3 m. Calculate its MA, VR and efficiency, and say which is more efficient, the crowbar or the ramp that has an efficiency of 62.5%. (d) A single fixed pulley has VR	(a) $MA = \text{load/effort} = 200/125 = 1.6$. $VR = \text{effort distance/load distance} = 16/8 = 2$. Efficiency = $(MA/VR) \times 100\% = (1.6/2) \times 100\% = 80\%$. (Check: useful work output 1600 J / work input 2000 J = 80%.) (b) With no friction, $MA = VR = 2$, so effort = $200 \text{ N} / 2 = 100 \text{ N}$. (c) $MA = 400/125 = 3.2$. $VR = 1.2/0.3 = 4$. Efficiency = $(3.2/4) \times 100\% = 80\%$. The crowbar (80%) is more efficient than the ramp (62.5%). (d) A single fixed pulley with $VR = 1$ only changes the direction of the force; the effort is still about 200 N. A child can pull downwards using body weight, but it does not reduce the force needed. The effort is reduced when the second, movable pulley is added.

= 1. What does it do, and why can a child still use it at the well?	
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Part 5: Final explanation	
Write a complete explanation (6 to 8 sentences) that answers the driving question: How do pulleys reduce the effort needed to lift water, and how does energy transfer in the process? Include: (i) the energy the bucket gains, (ii) the effort and the rope distance in the hand lift and in the pulley lift, (iii) where the energy comes from, (iv) what happens to the energy that does not become PE, and (v) the statement about conservation of energy. Use the numbers from the Kitui well.	The 20 kg bucket (200 N) gains 1600 J of potential energy when raised 8 m, whether Mama Kavata lifts it by hand or Mwende uses the pulley system, because the same weight is raised the same height. In the hand lift the effort is 200 N over 8 m, which is 1600 J of work. With the pulley system, Mwende pulls with only 125 N, but she pulls the rope 16 m, so she does 125 N x 16 m = 2000 J of work. The pulley does not give extra energy or reduce the work needed; it trades a smaller force for a longer distance, which is why the bucket feels lighter and a child can lift it. The energy comes from the chemical energy in her body from food such as ugali and githeri. Of the 2000 J put in, 1600 J becomes the PE of the bucket and 400 J (20%) is transferred as heat, and a little sound, by friction in the pulleys and rope, so the efficiency is 80%. Energy is conserved overall because none is created or destroyed; it is only transferred and transformed.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of data and calculations	All calculations (PE, work, power, KE, MA, VR, efficiency) are correct, use $g = 10 \text{ N/kg}$, show steps and give correct units.	Most calculations are correct with steps; one or two small slips in units or arithmetic.	Several calculations are missing, incorrect or have no units or working.
Explaining why the pulley feels easier	Clearly states a smaller force (125 N vs 200 N) over a longer distance (16 m vs 8 m) and that the work needed is not reduced.	States the smaller force and longer distance but is unclear about the work.	Says the pulley makes lifting easier without describing force and distance, or claims the pulley reduces work or gives energy.
Energy source, transfer and conservation	Identifies the chemical energy from food as the source, accounts for $2000 \text{ J} = 1600 \text{ J PE} + 400 \text{ J heat}$, and states that energy is conserved and not destroyed.	Identifies the energy source and the 400 J as heat, but conservation is incompletely stated.	Does not identify the source, or says the energy is lost or destroyed.
Machine terms and efficiency	Correctly defines and calculates MA, VR and efficiency, and compares machines and the fixed pulley accurately.	Calculates most of MA, VR and efficiency correctly with minor errors in comparison.	Confuses MA, VR and efficiency or cannot calculate them.

Scientific communication	The explanation is organised, uses scientific terms accurately and refers to the Kitui well data throughout.	The explanation is mostly clear with some terms used loosely.	The explanation is disorganised or uses few scientific terms.
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